WORKED SOLUTION

STEP I 2015, Question 8

Pure • Divisibility of Power Sums

STEP I 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q8

The key idea

Write the sum twice, once forwards and once backwards, and add: this is Gauss's trick for part (i), and it is exactly what part (ii) is built for when the terms are kth powers with k odd. Pairing r^k with $(n-r)^k$ shows that $2S$ is a multiple of n; pairing r^k with $(n+1-r)^k$ shows that $2S$ is also a multiple of $n+1$. Since n and $n+1$ share no common factor, the two facts combine to give divisibility by $\frac{1}{2}n(n+1)$.

Solution

Part (i)

Let $T = 1+2+\cdots+n$. Writing the sum in reverse order,

$$T = 1+2+\cdots+(n-1)+n,$$
$$T = n+(n-1)+\cdots+2+1.$$

Adding the columns, each of the n columns sums to $n+1$, so $2T = n(n+1)$ and

$$1+2+\cdots+n = \tfrac{1}{2}n(n+1).$$

Part (ii)

Working modulo N, we have $N-m \equiv -m$, so

$$(N-m)^k + m^k \equiv (-m)^k + m^k = -m^k + m^k = 0 \pmod{N},$$

where $(-m)^k = -m^k$ because k is odd. Hence N divides $(N-m)^k + m^k$.

If you prefer to avoid congruences: for odd k we have the factorisation

$$a^k + b^k = (a+b)(a^{k-1} - a^{k-2}b + a^{k-3}b^2 - \cdots + b^{k-1}),$$

(the signs alternate and there are k terms in the bracket, so the last sign is $+$). Taking $a = N-m$, $b = m$ gives $a+b = N$ times an integer. Nothing here needs $m \le N$: if $m > N$ then $N-m$ is negative, but it is still an integer.

Divisibility by n or $\frac{1}{2}n$

Let $S = 1^k + 2^k + \cdots + n^k$ with k odd. We copy the trick of part (i). Since $0^k = 0$, we may write

$$S = \sum_{r=0}^{n} r^k = \sum_{r=0}^{n} (n-r)^k,$$

the second form being the same sum in reverse order. Adding,

$$2S = \sum_{r=0}^{n} \Big[(n-r)^k + r^k \Big].$$

By part (ii) with $N = n$ and $m = r$, every bracket is divisible by n, so n divides $2S$. Say $2S = nM$ with M an integer.

- If n is odd, then n divides $2S$ and shares no factor with 2, so n divides S.
- If n is even, then $S = \frac{1}{2}n \cdot M$ with $\frac{1}{2}n$ an integer, so $\frac{1}{2}n$ divides S.

? **Ask yourself** Why can't we simply say "n divides S" when n is even?

Because the step from "$n \mid 2S$" to "$n \mid S$" needs n to be coprime to 2. Indeed it can fail: with $n = 2$, $k = 1$ we get $S = 3$, which is not a multiple of 2. All we can say for even n is that half of n divides S.

Divisibility by $1 + 2 + \cdots + n$

By part (i) we need $\frac{1}{2}n(n+1) \mid S$. Pair the terms differently: reversing $1, 2, \ldots, n$ sends r to $n + 1 - r$, so

$$2S = \sum_{r=1}^{n} \Big[(n+1-r)^k + r^k \Big],$$

and by part (ii) with $N = n + 1$ and $m = r$, every bracket is divisible by $n + 1$. So $n + 1$ divides $2S$.

Now n and $n + 1$ are consecutive integers, so they have no common factor other than 1; the same is then true of any factor of n and any factor of $n + 1$. We use the fact that if two coprime integers each divide S, then so does their product.

- *n odd.* We showed $n \mid S$. Also $n + 1$ is even and $n + 1 \mid 2S$, so writing $2S = (n+1)L$ gives $S = \frac{1}{2}(n+1)L$, i.e. $\frac{1}{2}(n+1) \mid S$. Since n and $\frac{1}{2}(n+1)$ are coprime, their product $\frac{1}{2}n(n+1)$ divides S.
- *n even.* We showed $\frac{1}{2}n \mid S$. Also $n + 1$ is odd and divides $2S$, so (being coprime to 2) it divides S. Since $\frac{1}{2}n$ and $n + 1$ are coprime, their product $\frac{1}{2}n(n+1)$ divides S.

In both cases S is divisible by $\frac{1}{2}n(n+1) = 1 + 2 + \cdots + n$, as required.

Ask yourself Why is it legitimate to combine two divisibility facts into one by multiplying the divisors?

If $a \mid S$ and $b \mid S$ with $\gcd(a, b) = 1$, write $S = aq$. Then $b \mid aq$ and b shares no prime factor with a, so every prime power dividing b must divide q; hence $b \mid q$ and $ab \mid S$. Without coprimality this fails: $4 \mid 12$ and $6 \mid 12$, but $24 \nmid 12$. That is why we were careful to pair n with $\frac{1}{2}(n+1)$, or $\frac{1}{2}n$ with $n+1$, rather than, say, n with $n+1$ when n is even.

Checks

Try $k = 3$, $n = 4$: $S = 1 + 8 + 27 + 64 = 100$, and $1 + 2 + 3 + 4 = 10$ divides 100. In fact $1^3 + \cdots + n^3 = \left(\frac{1}{2}n(n+1)\right)^2$, so for $k = 3$ the result is immediate. For $k = 5$, $n = 5$: $S = 1 + 32 + 243 + 1024 + 3125 = 4425 = 15 \times 295$. The final result was also confirmed by computer for every $n \leq 119$ and every odd $k \leq 15$. For even k the claim fails: $1^2 + 2^2 = 5$ is not divisible by 3, which shows that oddness of k is really used.

Common slips

- Concluding $n \mid S$ from $n \mid 2S$ when n is even.
- Multiplying divisors that are not coprime, for example claiming $n(n+1) \mid 2S$ and then halving without care.
- Forgetting the term $r = 0$ (or $r = n$) when pairing r^k with $(n-r)^k$: including $0^k = 0$ makes every term pair up neatly.

Beyond STEP

Power sums $1^k + \cdots + n^k$ are polynomials in n of degree $k+1$, given by Faulhaber's formula in terms of the Bernoulli numbers; for odd k they can even be written as polynomials in $\frac{1}{2}n(n+1)$, which explains the result from another angle. The argument here is a first taste of working with congruences and coprimality, the bread and butter of elementary number theory. Davenport's *The Higher Arithmetic* is a lovely introduction.

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